Diabetic Retinopathy Image Segmentation Using Deep Learning Models

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Abstract— Diabetic retinopathy (DR) is a prominent global cause of vision loss, underscoring the need for accurate and timely diagnosis to ensure effective management. This study tackles the challenge of DR classification by leveraging deep learning (DL) models. We employed a comprehensive dataset from Kaggle, which includes retinal images annotated with DR severity levels. The preprocessing phase involved image normalization and the application of the Synthetic Minority Over-sampling Technique (SMOTE) to bolster model training.

We trained and evaluated three DL models: Convolutional Neural Network (CNN), RESNET50, and EFFICIENTNET-B2 assessing their proficiency in classifying DR severity levels. Among these, EFFICIENTNET-B2 demonstrated superior performance, achieving an accuracy of 77% across five DR classes. To enhance accuracy further, EFFICIENTNET-B2 was fine-tuned using various optimization functions, with the Adam optimizer producing the most favorable results. Analyses of specificity and sensitivity provided insights into each model's diagnostic strengths and limitations.

***Index terms*— Diabetic Retinopathy, Deep Learning, RESNET50, EFFICIENTNET-B2, Medical Imaging**

I. INTRODUCTION

A. Background

Diabetic retinopathy (DR) is a severe complication of diabetes mellitus and a leading cause of blindness worldwide [1]. DR affects the retina, a light-sensitive tissue at the back of the eye, and is characterized by damage to blood vessels due to high blood sugar levels [2].

Over time, these vessels may leak blood or fluid, leading to swelling and forming deposits known as exudates. In advanced stages, new abnormal blood vessels can develop on the retina's surface, potentially leading to severe vision loss or blindness if left untreated. Without timely management, DR can result in permanent blindness.

B. Importance of Early Detection

Early detection and treatment are crucial in effectively managing diabetic retinopathy. Regular screening and prompt action can prevent or delay vision loss in many individuals, offering hope and motivation to those with diabetes [3]. Digital retinal imaging has modernized the diagnosis and monitoring of DR, allowing healthcare providers to detect complex changes in the retina that may indicate the onset or progression of the disease. Automated image analysis techniques, particularly image segmentation, hold immense potential in extracting relevant features from retinal images to help clinicians make accurate diagnoses and plan treatments. These techniques are poised to play an increasingly essential role in diabetic retinopathy management.

C. Contribution

This study makes several critical contributions to diabetic retinopathy (DR) diagnosis using deep learning (DL) techniques. We integrated multiple advanced DL models, including Convolutional Neural Network (CNN), RESNET50, and EFFICIENTNET-B2, into a unified framework for DR classification. This comprehensive framework, designed for robustness and reliability, instills confidence in its effectiveness for medical professionals, researchers, and developers. Our study demonstrates the effectiveness of fine-tuning and optimization functions in improving model performance. Among the evaluated models, EFFICIENTNET-B2 outperformed CNN and RESNET50, achieving an accuracy of 77% across five DR classes. This improvement in performance underscores the effectiveness of our approach. Extensive specificity and sensitivity analyses were conducted to identify each model's diagnostic strengths and weaknesses. These analyses provide valuable insights into the practical applicability of the models in clinical settings, making our research directly relevant to medical professionals. The classification system, implemented using Python Flask, provides a robust real-time image upload and DR prediction platform through a user-friendly interface. This integrated approach not only highlights the practical benefits of advanced DL models in enhancing DR diagnosis and management but also instills optimism about the future of medical imaging and diagnostic accuracy, ultimately improving patient outcomes.

II. RELATED WORK

This literature review will discuss the recent advancements in Diabetic retinopathy detection with the help of deep learning and image processing techniques, highlighting the methodologies, challenges, and results from various studies.

In 2018, Burewar et al. [4] proposed a method for detecting and classifying different stages of diabetic retinopathy using a combination of U-Net segmentation with region combining and CNN techniques applied to high-resolution retinal fundus images. The method is developed to diagnose and classify images based on the severity of the disease, explicitly addressing the significant challenge of variability, particularly in cases of proliferative diabetic retinopathy characterized by retinal proliferation and detachment. The proposed method focused on accurate retinal vessel segmentation to enhance classification precision. Through region merging, the method ensures that features lost during segmentation are retained and effectively utilized by the image classifier. Results showed an accuracy of up to 93.33%, demonstrating good outcomes for automated diabetic retinopathy diagnosis and classification.

This study by Deperlıoğlu and Köse (2018) [5] presents an innovative approach for diagnosing diabetic retinopathy utilizing image processing and deep learning methodologies on retinal fundus images. The research incorporates a sophisticated image enhancement process involving HSV transformation, V transform algorithm, histogram equalization, and Gaussian low-pass filtering. After these enhancements, image classification is executed using a Convolutional Neural Network (CNN). The methodology is assessed using 400 retinal fundus images sourced from the Kaggle Diabetic Retinopathy Detection database, with classification performance evaluated at various stages of image processing. Each stage involves conducting twenty experiments, from which average values are derived. The study reports exceptional performance metrics, including an accuracy of 97%, sensitivity of 96.67%, specificity of 93.33%, precision of 97.78%, F-score of 93.33%, and recall of 93.33%. These results underscore the efficacy of the proposed method in accurately diagnosing diabetic retinopathy, highlighting its potential for clinical application in the detection and management of this ocular disease.

Further advancing the field, Hemanth et al. (2020) [6] propose a hybrid method for diagnosing diabetic problems from retinal fundus images, combining image processing and deep learning to enhance diagnostic accuracy. Recognizing the complexity of reliable diabetic retinopathy detection, the study aims to reduce manual interpretation efforts and enhance diagnostic performance through automated systems. The proposed method employs image processing techniques, including histograms, to preprocess the images, followed by classification using a Convolutional Neural Network (CNN). Validation is conducted using approximately 400 retinal fundus images from the MESSIDOR database, achieving performance metrics of 94% sensitivity (recall), 97% accuracy, 94% precision, 98% specificity, and 94% F-score. These results highlight the method's effectiveness in providing a reliable automated diagnostic tool for diabetic retinopathy.

Adding to the advancements, Da Rocha et al. (2022) [7] propose using the VGG16 neural network to classify diabetic retinopathy into five categories. The study addresses common issues in retinal images, such as inadequate lighting, low contrast, and noise levels that can hinder accurate medical analysis. The methodology involves preprocessing steps such as resizing images, cleaning low-quality data by reclassifying them into class 5, data augmentation, and balancing classes during training. Hyperparameter tuning optimizes the VGG16 model for image classification. Results have shown superior performance on the DDR database compared to existing methods, achieving high accuracy and robust metrics. This research advances state-of-the-art diabetic retinopathy diagnosis, particularly in handling diverse image qualities and enhancing diagnostic outcomes across multiple datasets.

Deshpande and Pardhi (2021) [8] propose using a pre-trained CNN, VGG16, to detect the severity of diabetic retinopathy from retinal images. Using the Asia-Pacific Tele-Ophthalmology Society (APTOS) 2019 Blindness Detection dataset, which contains 3668 retinal images captured through fundus photography, the model is trained and tested on a separate set of 1728 images not included in the training dataset. The approach includes using the Categorical Cross-Entropy loss function to minimize loss and the ADAM optimizer to enhance training speed and efficiency. The model outputs an integral value on a scale of 0-4, indicating the severity of diabetic retinopathy. The main challenge addressed is achieving reliable accuracy in classifying the severity of the disease. The model achieved an accuracy rate of 74.58% over 30 epochs, with a loss of 5.06%. This model aims to assist doctors in the early detection of diabetic retinopathy, contributing to timely medical interventions.

Additionally, Aatila et al. (2021) [9] proposed an intelligent system for diabetic retinopathy (DR) classification using deep learning (DL) tools, primarily CNNs, to assist ophthalmologists in making preliminary diagnoses. The system categorizes retinal images into five classes: normal eyes, mild DR, moderate DR, severe DR, and proliferative DR. The proposed approach employs VGG-16 and ResNet50 models to classify DR using color retinal background images. The main challenge addressed is achieving reliable classification accuracy to support early diagnosis and intervention. The results showed that the VGG-16 model achieves a classification accuracy of 70%, while the ResNet50 model achieves an accuracy of 25%. This system aims to enhance the screening process for DR, providing timely and adequate medical responses to this leading cause of blindness.

Wang et al. (2018) [10] proposed using deep CNN methods for the stage classification of diabetic retinopathy (DR), an essential step in evaluating and managing the disease. Due to high blood glucose levels causing damage to retinal blood vessels, various microstructures, such as microaneurysms, hard exudates, and neovascularization, can

appear in the retina. This work categorizes representative DR images into five classes based on ophthalmologists' expertise. The approach includes using multiple deep CNN methods to classify these stages, with InceptionNet V3 achieving state-of-the-art accuracy. The main challenge addressed is the reliable recognition and classification of different DR stages using retinal images. The success of InceptionNet V3 in this task shows the performance of deep CNNs in biomedical image analysis and DR stage classification, contributing to improved diagnostic accuracy and patient management.

Saranya et al. (2022) [11] proposes a model to detect the stages of diabetic retinopathy, which accurately identifies neovascularization characterized by the growth of abnormal blood vessels found in the retina. Recognizing the challenges given by traditional medical tests, which are time-consuming and expensive, the study mainly focuses on using image processing and deep learning techniques to address this issue well. The proposed system processes color fundus images through steps such as resizing. For classification, the model has used different layers of a CNN combined into a VGG-16 architecture. Trained and tested on around 2200 images taken from the Kaggle database, the model has shown its performance on the DRIVE and STARE datasets, achieving high-performance metrics: 0.99 specificity, 0.96 accuracy, etc.

III. METHODOLOGY

This section will provide an overview of all three models used in this study.

A. CNN

CNNs play an essential role in this study for diabetic retinopathy (DR) classification because they can extract complex hierarchical features from images [12]. In medical imaging, CNNs are good at learning spatial hierarchies of features, which is essential for having good patterns of different diabetic retinopathy seriousness levels. The CNN architecture used in this study has multiple types of convolutional layers, which are, of course, followed by pooling layers to progressively extract and downsample features from retinal images [13]. These layers are designed to capture low-level features like edges and textures and high-level features such as lesions and microaneurysms characteristic of DR. Batch normalization, and dropout layers are combined to enhance model generalization and mitigate overfitting, respectively. Training the CNN includes optimizing classification accuracy across five classes of diabetic retinopathy seriousness using a vast dataset. Transfer learning techniques using pre-trained models on large datasets, like ImageNet, were also explored to expedite convergence and improve performance.

Ultimately, CNN's ability to learn discriminative features from retinal images helps in robust diabetic retinopathy seriousness classification, which enhances diagnostic accuracy and clinical decision-making.

B. ResNet 50

ResNet50, a sophisticated variant of Convolutional Neural Networks (CNNs), is innovatively employed in this study to classify diabetic retinopathy (DR). Its profound architectural depth and the implementation of residual learning techniques make it a promising tool for medical image analysis. The network's design, notably its incorporation of residual blocks, significantly enhances the training of profound networks by effectively addressing and mitigating the vanishing gradient problem [14].

This attribute is especially beneficial in medical image analysis, where the ability to capture and discern intricate features across various levels of DR severity is essential. This methodology, ResNet50, extracts hierarchical features from retinal images, beginning with convolutional layers that identify basic image features and progressively transitioning to deeper layers that capture more complex patterns. To further refine the feature extraction process, max pooling, and global average pooling layers are employed for spatial down-sampling and feature aggregation [15].

Additionally, the model integrates batch normalization and dropout layers to enhance overall stability and minimize the risk of overfitting during training. To maximize the model's performance with a limited amount of labeled medical data, transfer learning is applied by initializing ResNet50 with weights pre-trained on the extensive ImageNet dataset [16]. Subsequently, the model is fine-tuned on the DR dataset to tailor its learned features specifically for the classification of diabetic retinopathy.

ResNet50's robust capability to manage deep networks and effectively learn discriminative features is crucial in accurately classifying the seriousness or severity of diabetic retinopathy in this study. This robustness ensures that the model identifies and correctly categorizes the varying levels of DR, thereby significantly enhancing diagnostic accuracy and supporting clinical decision-making. This reliability of ResNet50 should provide reassurance about the model's performance in medical image analysis.

C. EfficientNetB2

EfficientNet-B2 is used in this study for diabetic retinopathy (DR) classification due to its good performance and scalability in deep learning architectures. The model's scaling method optimizes network depth, width, and resolution simultaneously [17], achieving state-of-the-art (SOTA) performance with fewer parameters than traditional CNNs.

In this methodology, EfficientNet-B2 begins with convolutional type layers designed to extract features from retinal images, followed by depthwise-separable convolution layers to capture spatial things and channel-wise dependencies in a good way. The model's efficient architecture enables it to handle high and prominent features [18] across different diabetic retinopathy seriousness levels. Transfer learning plays a vital role by initializing EfficientNet-B2 with weights pre-trained on large-scale image datasets like ImageNet, enhancing its ability to generalize and extract relevant features from medical images with limited labeled data.

IV. EMPIRICAL STUDY

A. Experimental Setup

In this experimental setup, we have used a good array of libraries to generate and do the preprocessing, augmentation, and analysis of the retinal images for Diabetic Retinopathy and diabetic retinopathy seriousness classification. The primary libraries included OpenCV for image processing, enabling us to resize and preprocess images well. TensorFlow and Keras were employed to build and train deep learning models, using EfficientNetB2 for its advanced architecture. Data manipulation and analysis were conducted using Pandas and NumPy, while visualization tasks were handled using Matplotlib and Seaborn. The scikit-learn library was essential for different machine learning tasks, including train-test splitting and performance evaluation through metrics like confusion matrix and classification report. We used the SMOTE technique from the imbalanced-learn (learn) library to handle a class imbalance in the dataset. ImageDataGenerator from Keras gave data augmentation, enhancing the robustness of the training set by applying random transformations. The experimental workflow also included callbacks from TensorFlow, such as EarlyStopping and ReduceLROnPlateau, to optimize model training. Also, TSNE from sci-kit-learn was utilized to visualize the high-dimensional data types, and PrettyTable was used for structured output representation. The combination of these vast libraries ensured a robust and effective experimental setup, allowing for the effective preprocessing, training, and evaluation of deep learning models in the diabetic retinopathy classification.

B. Dataset Description

The dataset used in this study comes from the Kaggle APTOS 2019 Blindness Detection competition. It is a collection of retina images captured with the help of fundus photography under a wide array of imaging conditions. Each image is rated by clinicians for the seriousness of Diabetic Retinopathy on a scale from 0 to 4, where 0 shows no diabetic retinopathy and four shows Proliferative diabetic retinopathy. This categorization gives an excellent analysis of diabetic retinopathy levels. Real-world data challenges are evident in this dataset, including images with noise, artifacts, and different degrees of focus, exposure, and illumination. These images were sourced from multiple clinics using diverse camera equipment, contributing to the dataset's inherent variability. The dataset's diversity and complexity make it an ideal candidate for training robust deep-learning models capable of accurately classifying diabetic retinopathy. Proper preprocessing techniques, such as image resizing and noise reduction, are essential to handle these variations and enhance model performance.

C. Data Preprocessing

Data preprocessing is crucial for ensuring the deep learning models effectively classify the retinal images for Diabetic Retinopathy (DR) seriousness. The initial preprocessing steps, which include importing necessary libraries, loading the dataset, and labeling the images with their respective diabetic retinopathy levels, play a pivotal role in this process. These steps set the foundation for the subsequent preprocessing tasks. Images are then resized uniformly, typically using OpenCV's resizing functions. To handle the variability and noise in the dataset, we convert color images to grayscale when necessary, using a custom function conv_gray. The crop_image_from_gray function plays a crucial role in enhancing the focus on the retina by removing unnecessary background, thereby improving the quality of the input data for the model. This function identifies and retains the region of interest based on pixel intensity thresholds. Images undergo circular cropping to standardize the dataset further, ensuring consistent focus around the retina's center. This step uses the circle_crop function, which creates a circular mask and applies Gaussian blurring to smoothen the image.

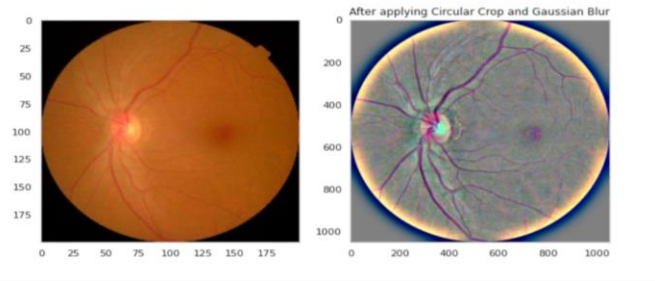

Fig 1: Comparison of Original Retinal Image and Image After Applying Circular Crop and Gaussian Blur

D. Data Splitting

Splitting the dataset into training and testing sets is crucial to ensure that the deep learning models are trained well and evaluated accurately. The dataset, which contains retinal images labeled with their respective Diabetic Retinopathy (DR) seriousness levels, is first preprocessed to standardize image sizes and enhance the focus on the region of interest. Once preprocessing is complete, the dataset is divided into two main subsets: the training and testing sets. This split is mainly done in an 80:20 ratio, where 80% of the data is allocated for training the model, and 20% is reserved for testing its performance—the train_test_split function from the sklearn.model_selection module is used to ensure a random and unbiased data division. To maintain a balanced representation of all diabetic retinopathy seriousness levels across both subsets, this technique ensures that each seriousness level (0-4) is proportionally represented in the training and testing sets, which is very important for the model to learn and generalize well across all classes. The training set is then augmented using rotation, flipping, and scaling techniques to increase the training data's diversity, enhancing the model's robustness and ability to generalize to unseen data. The robustness of the model is further enhanced using the ImageDataGenerator from Keras, which generates augmented images on the fly during model training, ensuring that the model is well-prepared to handle unseen data.

This architecture diagram in Fig. 2 illustrates the process of utilizing the APTOS 2019 Blindness Detection Dataset for diabetic retinopathy classification. Initially, the raw images undergo a preprocessing phase, which includes image resizing and other preprocessing steps. Subsequently, the data is split into training and testing sets. This study employs a diverse range of models, including CNN, ResNet50, and EfficientNet-B2, each contributing to the comprehensive nature of the research. These models are trained on the processed data. After the training phase, the models are evaluated for their performance. Finally, the trained models are used to predict diabetic retinopathy.

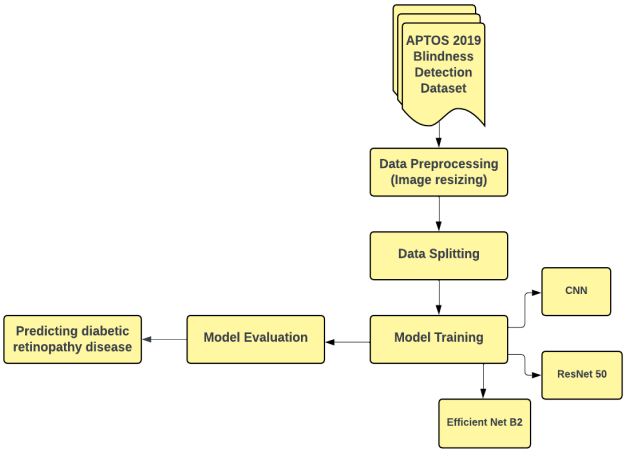

Fig 2: Workflow for Diabetic Retinopathy Detection Using Deep Learning Models

V. EVALUATION RESULT

A. Classification Performance of CNN Model

The CNN model exhibits varied performance when classifying different severity levels of Diabetic Retinopathy (DR) on the test dataset, achieving an overall accuracy of 72%. Detailed precision, recall, and F1-score analysis reveal significant variability across DR classes.

Class 0 (No DR) is where the model truly shines, boasting a precision of 86% and a recall of 99%. These high figures underscore the model's reliability in correctly identifying the absence of diabetic retinopathy. However, for Class 1 (Mild DR), the precision drops to 73% and recall to 14%, indicating the model's struggle in detecting mild cases. Class 2 (Moderate DR) shows moderate performance with a precision of 52% and recall of 77%. Performance deteriorates further for Class 3 (Severe DR) and Class 4 (Proliferative DR), with significantly lower precision and recall.

Overall, the CNN model's commendable accuracy provides a reassuring foundation, despite its inconsistent performance across different DR severity levels. This variability underscores the urgent need for further refinement to improve diagnostic accuracy across all classes of diabetic retinopathy.

The classification performance of the CNN model for diabetic retinopathy (DR) severity is detailed in Table I. The performance metrics include Precision, Recall, and F1-Score for each DR class.

TABLE I
CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score
No DR	0.86	0.99	0.92
Mild	0.73	0.14	0.23
Moderate	0.52	0.77	0.62
Severe	0.50	0.03	0.05
Proliferative DR	0.33	0.07	0.11

The sensitivity and specificity metrics for the CNN model across different classes of diabetic retinopathy (DR) severity are detailed in Table II.

TABLE II
SENSITIVITY AND SPECIFICITY OF CNN MODEL

Class	Sensitivity	Specificity
No DR	0.842391	0.989041
Mild	0.993865	0.135802
Moderate	0.746765	0.770833
Severe	0.995863	0.027027
Proliferative DR	0.988148	0.068966

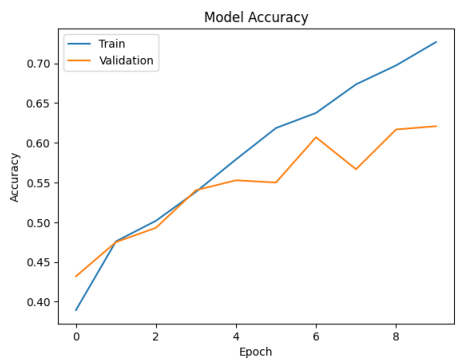

Fig 3: Accuracy Graph of CNN model

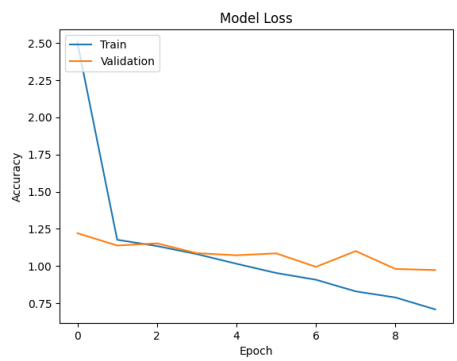

Fig 4: Loss Graph of CNN Model

B. *Classification Performance of ResNet 50 Model*

The RESNET50 model exhibits varied performance across the different severity levels of Diabetic Retinopathy (DR), as evaluated on the test dataset. The model's overall accuracy is reported at 75%, indicating its capability to classify DR severity with reasonable accuracy across multiple classes. Class 0 (No DR) shows strong performance with high precision (92%) and recall (98%), indicating robust identification of images without DR. Class 1 (Mild DR) demonstrates moderate precision (50%) and recall (49%), suggesting challenges in distinguishing mild cases accurately. Class 2 (Moderate DR) exhibits reasonable precision (62%) and recall (72%), showing improved performance compared to the CNN model in identifying moderate cases. However, Class 3 (Severe DR) and Class 4 (Proliferative DR) display lower precision and recall scores, indicating significant difficulty in correctly classifying severe and proliferative cases. Class 4, in particular, shows a high precision (100%) but very low recall (2%), indicating that while the model rarely misclassifies non-proliferative cases, it needs help correctly identifying proliferative cases. So, while RESNET50 improves upon the CNN model's performance in certain aspects, particularly with higher precision in some classes, challenges remain in achieving balanced recall across all severity levels.

The classification performance of the ResNet50 model for diabetic retinopathy (DR) severity is detailed in Table III. The performance metrics include Precision, Recall, and F1-Score for each DR class.

TABLE III
CLASSIFICATION REPORT FOR RESNET50 MODEL

Class	Precision	Recall	F1-Score
No DR	0.92	0.98	0.95
Mild	0.50	0.49	0.50
Moderate	0.62	0.72	0.67
Severe	0.36	0.35	0.36
Proliferative DR	1.00	0.02	0.03

Table IV illustrates that the sensitivity and specificity values for the ResNet50 model vary across different classes of DR severity.

TABLE IV
SENSITIVITY AND SPECIFICITY OF RESNET 50 MODEL

Class	Sensitivity	Specificity
No DR	0.910326	0.983561
Mild	0.938650	0.493827
Moderate	0.842884	0.723958
Severe	0.966954	0.351351
Proliferative DR	1.000000	0.017241

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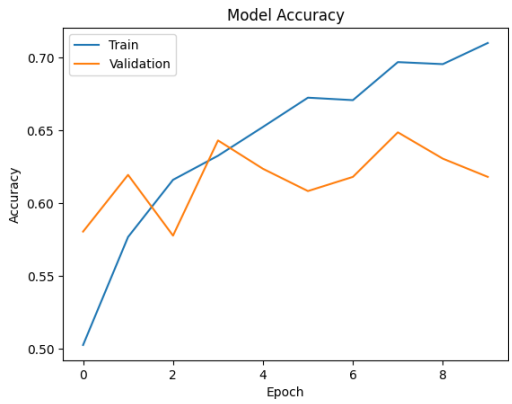

Fig 5: Accuracy Graph of ResNet 50 model

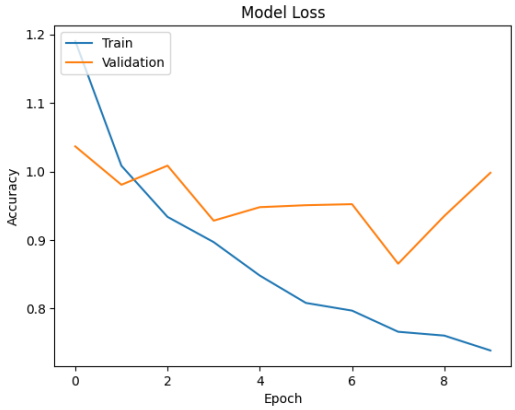

Fig 6: Loss Graph of ResNet 50 Model

C. *Classification Performance of EFFICIENTNET-B2 Model*

EFFICIENTNET-B2 demonstrates the highest accuracy among the evaluated models, achieving an overall accuracy of 77% on the test dataset for Diabetic Retinopathy (DR) severity classification. The model exhibits varying precision, recall, and F1-score metrics across different DR severity classes. Class 0 (No DR) shows excellent performance with high precision (93%) and recall (99%), indicating robust identification of images without DR. Class 1 (Mild DR) exhibits improved precision (70%) compared to other models, though with moderate recall (32%), indicating some challenges in correctly identifying mild cases. Class 2 (Moderate DR) demonstrates reasonable precision (64%) and good recall (81%), showing effective performance in identifying moderate cases.

However, Class 3 (Severe DR) and Class 4 (Proliferative DR) display lower precision and recall, indicating difficulties in accurately classifying severe and proliferative cases. Class 4, in particular, shows the lowest performance metrics with precision (50%) and recall (17%), highlighting challenges in correctly identifying proliferative cases.

The classification performance of the EfficientNet-B2 model for diabetic retinopathy (DR) severity is detailed in Table III. The performance metrics include Precision, Recall, and F1-Score for each DR class.

TABLE V
CLASSIFICATION REPORT FOR EFFICIENTNET-B2 MODEL

Class	Precision	Recall	F1-Score
No DR	0.93	0.99	0.96
Mild	0.70	0.32	0.44
Moderate	0.64	0.81	0.71
Severe	0.34	0.38	0.36
Proliferative DR	0.50	0.17	0.26

TABLE V1
SENSITIVITY AND SPECIFICITY OF EFFICIENTNET-B2 MODEL

Class	Sensitivity	Specificity
No DR	0.923913	0.994521
Mild	0.983129	0.320988
Moderate	0.835490	0.807292
Severe	0.961207	0.378378
Proliferative DR	0.985185	0.172414

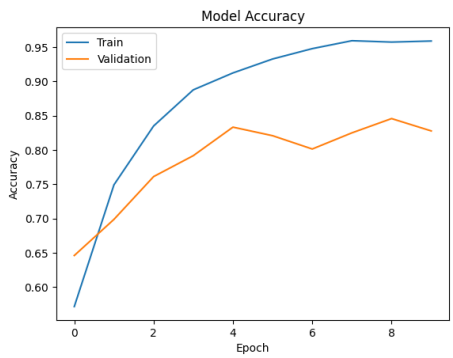

Fig 7: Accuracy Graph of Efficientnet-B2 model

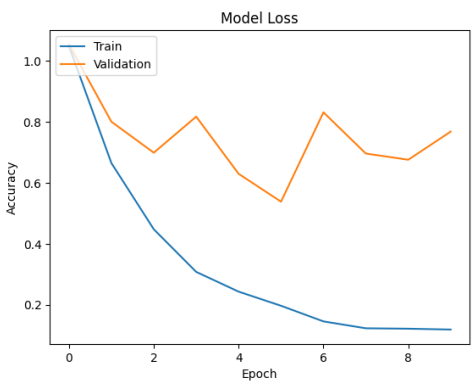

Fig 8: Loss Graph of Efficientnet-B2 Model

The accuracy performance of the three models used in this study for diabetic retinopathy (DR) classification is detailed in Table VII.

TABLE VII
MODEL ACCURACIES

Model	Accuracy
CNN	72%
ResNet 50	75%
Efficient Net B2	77%

Among the models evaluated, EfficientNet-B2 achieved the highest accuracy at 77%. This model outperformed the CNN and ResNet50 models, which achieved accuracies of 72% and 75%, respectively. The superior performance of EfficientNet-B2 can be attributed to its advanced architecture and the application of fine-tuning techniques. Specifically, the Adam optimizer was employed to enhance the model's performance by optimizing the training process through various optimization functions. This fine-tuning approach contributed to EfficientNet-B2's robust classification accuracy in detecting and classifying the severity of diabetic retinopathy.

VI. CONCLUSION & FUTURE SCOPE

This study effectively addressed the challenge of classifying diabetic retinopathy (DR) using advanced deep-learning models. The dataset sourced from the Kaggle APTOS 2019 Blindness Detection competition provided a substantial collection of retinal images with varying DR degrees, enabling robust model training and evaluation. Preprocessing steps, including image resizing and normalization, ensured data suitability for the CNN, ResNet50, and EfficientNet-B2 models. The SMOTE oversampling technique was employed to manage class imbalance effectively. EfficientNet-B2 emerged as the most accurate model, achieving an overall classification accuracy of 77%, outperforming the CNN and ResNet50 models. The CNN model excelled in identifying the absence of DR, while ResNet50 and EfficientNet-B2 showed superior performance in detecting moderate DR cases. Sensitivity and specificity analyses highlighted each model's strengths and areas for improvement. EfficientNet-B2 was fine-tuned using various optimization functions, with the Adam optimizer selected for its effectiveness.

Future work in DR classification could benefit from incorporating advanced deep learning architectures, such as DenseNet, Inception, or Xception, to improve accuracy by capturing more intricate features from retinal images. These models have demonstrated exceptional performance in image recognition tasks and could significantly enhance DR severity classification.

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